rogii-wellbore-geology-prediction

9th Place Solution

Rank	9
Team	tremors
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Version	Original

Source: https://www.kaggle.com/competitions/rogii-wellbore-geology-prediction/discussion/733150
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First of all, thanks to ROGII and Kaggle for a genuinely unusual competition. And 6K competitors? Whoa. Also congrats to my teammate @datafan07 for the Competition GM title!

Solution TL;DR: Treat the well as an image over (candidate TVT × measured depth), let CNNs assign a categorical TVT distribution per MD column, average everything over 3×5 seeded folds, and blend with position-dependent weights. Trust honest CV over the public leaderboard.

The topic is incomplete, we will be adding details.

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1. Autoregressive Approach

This model treats tracking as repeated local classification over candidate offsets.

Input image: For each chunk of the lateral, we build a cost-volume image of shape `[9, 64 + 120, 41]`: 64 look-back context rows plus a 120-row chunk (MD pooled ×3), against **41 candidate offset lanes spanning ±30 ft** around the current anchor. The 9 channels encode GR-vs-typewell match costs and geometry for each candidate lane.

Architecture: 8 dilated `Conv2d` blocks (64 channels, GroupNorm + GELU, dropout 0.05) followed by a 2-layer BiGRU over the MD axis. Softmax over the 41 lanes, they resemble expected offset per row.

- At inference the model runs autoregressively: predict a chunk, re-anchor on its own output, predict the next chunk.
- Training samples random `(well, start position, anchor noise, prior walk)` chunks so the training distribution matches the inference rollout distribution, with a soft-CE + Huber loss.
- At inference we average 6 rollout configurations (chunk length ∈ {100, 120, 140} × phase ∈ {0, 40}).

Member OOF: 7.38 ft. Not our strongest member, but persistently non-zero in every blend fit.

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2. Whole-well U-Nets

The strongest members reframe the whole problem as dense segmentation:

- Each well becomes a single image. The vertical axis is candidate TVT relative to the anchor (384 bins × 0.75 ft ≈ ±144 ft); the horizontal axis is the entire lateral compressed in MD (640 columns, ×16 compression, plus 128 history rows).
- The network assigns one categorical TVT distribution to every MD column.

Architecture: A ResNet-34-encoder U-Net, column-wise softmax over the 384 depth bins. Input channels carry the GR-vs-typewell cost surface (signed and absolute), trajectory geometry, and the known-TVT history mask. Decoding takes the per-column expected row, anchored at the last known TVT.

We introduced two variants for diversity:

- **well-unet:** trained with vertical-shift jitter and horizontal-flip augmentation on the stock image.

- **relief-unet:** same architecture, but with two geometry-warping augmentations applied per sample:
  - **path-relief scaling:** Rescale the relief of the target path, re-render the image around the new path with the residual preserved
  - **thickness-warp:** A smooth monotone warp of the typewell depth axis, simulating thickness misfit between typewell and well location. The cost channels are recomputed after each warp, so the network trains on physically-consistent perturbed wells rather than pixel noise.

The two U-Nets ended almost identical in RMSE but decorrelated.

2.1 Entropy-weighted TTA

Inference uses 10 views per well: 5 vertical shifts (0, ±12, ±24 bins) × horizontal flip. Stock pooling sums the 10 posteriors equally.

We replaced it with:

- Compute each view's mean column entropy, then weight views by `softmax(−entropy / T)`.
- Temperature was tuned per model on OOF (`T = 10` for wellunet, `T = 5` for reliefunet).

Member	stock TTA	entropy TTA
wellunet OOF	6.33	**5.86**
reliefunet OOF	~6.0	**5.86**

We also utilized two physics-based approaches, @datafan07 will explain them under here.

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3. Seed-structured CV and OOF construction

The whole artifact is seed-structured:

- **3 fold seeds × 5 GroupKFold folds**
- **Test** prediction = mean over all 15 checkpoints.

Averaging fold seeds was worth roughly −0.15 pooled RMSE over any single seed.

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4. Stack Layer: Banded NNLS + Spread Calibration

Member reliability seemed position-dependent: Trackers were strong near the anchor, whole-well models are strongest mid-lateral and at the toe. So we have tried a **banded** combiner:

1. Splits rows into 7 quantile bands of `md_since_anchor`. Per band, fit a **no-intercept, non-negative least squares** of the target delta on the raw member deltas.
2. Computes each row's member std across member deltas, split into 80 quantile bins, and fit a per-bin multiplicative correction `m = ⟨p, y⟩ / ⟨p, p⟩` on OOF. It aims to shrink the blend where members disagree.

Each band picks a different specialist model:

band (heel→toe)	segnet	u-net	relief-unet	matcher-family
0 (heel)	0.20	0.09	0.02	high
3 (mid)	0.07	0.26	0.02	med
6 (toe)	0.14	0.00	**0.64**	low

The relief-trained U-Net owning the toe where thickness misfit accumulates is exactly what its augmentation was designed for. 👀

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5. Validation

We tracked three numbers for every candidate:

- chain: in-sample pooled OOF RMSE of the fitted combiner.
- heldfold: refit the **entire combiner** on 4/5 folds, score the held fold, pool.
- strict per-foldseed: heldfold where fold seed `s` uses **only fold-seed-`s` models and boundaries**

Final candidates (pooled RMSE):

Combiner	chain	heldfold	strict (mean over 3 fold seeds)	Public LB	Private LB
plain 4-member NNLS	5.42	5.47	5.56	**5.129**	6.503
band7+cal80, 8 members	**5.06**	**5.27**	**5.28**	5.435	**6.251**

Thanks for reading, and congratulations to the winners!